SEAT NO. CT-23025

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR (Computer Science)

FALL SEMESTER EXAMINATIONS 2024

Time : 3 Hours | **Batch 2023** | **Dated : 12-DEC-24**

Max Marks : 60

Probability & Statistics - MT-331

INSTRUCTIONS:

- **Attempt All Questions.**
- **Statistical Annexure is attached with Question paper.**

CLO—1

Q.1 The following data represents the response times (ms) for database queries **(12)**

Response Times (ms) for Database Queries									
43	91	13	76	28	119	67	85	32	58
41	98	19	74	29	88	61	46	102	35
79	55	93	25	68	82	39	59	96	14
71	86	48	104	31	77	62	89	42	99
26	66	80	54	94	38	70	84	49	110

$\bar{x} = 63.9$
$\sigma = 27.3$
$S = 27.581$

(a) Summarize the data of response time (ms) in a frequency distribution table using class intervals 0 – 19, 20 – 39 ... 100 – 119.

(b) Calculate average, median, mode and standard deviation of response time.

(c) Construct box plot and Identify outliers.

CLO—2

Q.2 (a) A software company analyzes customer feedback on its mobile app. The feedback is categorized as positive or negative, and the customers are classified as power users, regular users or casual users. **(06)**

	Power Users	Regular Users	Casual Users
Positive Feedback	120	80	40
Negative Feedback	20	30	60

Find the probability that a user is:

1. Power user given that they have provided positive feedback.
2. Not a casual user given that they have provided negative feedback.

(b) A regional telephone company operates three identical relay stations at different locations. During a one-year period, the number of malfunctions reported by each station and the causes are shown below. **(06)**

Station	A	B	C
Problems with electricity supplied	2	1	1
Computer malfunction	4	3	2
Malfunctioning electrical equipment	5	4	2
Caused by other human errors	7	7	5

Suppose that a malfunction was reported and it was found to be caused by other human errors. What is the probability that it came from station *C* ?

P.T.O

Q.3 **(a)** In testing a certain kind of truck tire over rugged terrain, it is found that 25% of the trucks fail to complete the test run without a blowout. Of the next 15 trucks tested, find the probability that: **(04)**

1. From 3 to 6 have blowouts.
2. Fewer than 4 have blowouts.

(b) A database experiences an average of 3.5 transactions/min. Find the probability that: **(04)**

1. At most 5 transactions occur in the next minute.
2. More than 7 transactions occur in the next 2 minutes.

(c) Network latency refers to the delay between sending and receiving data; follows an exponential distribution with the following probability density function: **(04)**

$$f(x) = \frac{1}{2}\, e^{-x/2} \quad , x > 0$$

1. Find cumulative distribution function of x.
2. Probability that latency exceeds 2 ms.

Q.4 **(a)** A cloud computing provider analyzes the storage capacity usage of its customers. The usage is normally distributed with a mean (μ) of 500 GB and a standard deviation (σ) of 100 GB. Find **(06)**

1. P (Customers use more than 700 GB)
2. P (Customers use between 300 GB and 600 GB)
3. 99% of the customers use less than or equal to what storage capacity.

(b) The life, in years, of a certain type of electrical switch has an exponential distribution with an average life β = 2. If 100 of these switches are installed in different systems, what is the probability that at most 35 fail during the first year? **(06)**

Q.5 **(a)** A cyber security firm claims that the average time taken to detect malware is 10 seconds. A random sample of 12 detection times (in seconds) is collected to test this claim. **(06)**

Detection times (in seconds)					
9.2	10.5	8.8	10.8	9.5	11.1
10.2	9.8	8.5	10.9	9.1	10.6

$\bar{x} = 9.9167$
$S = 0.887$

Test the hypothesis at 5% level of significance that the average time taken to detect malware is 10 seconds and also construct 95% confidence interval for average time taken to detect malware.

(b) A gaming company investigates the relationship between the processor speed (X, in GHz) and frames per second (Y) in a new game. **(06)**

X	2.5	3.2	2.8	3.5	2.2	3.0	2.6	3.8
Y	60	80	70	90	50	75	65	100

Calculate the linear regression model $\hat{y} = b_0 + b_1 x$ and quadratic regression model $\hat{y} = b_0 + b_1 x + b_2 x^2$ and also find coefficient of determination (R^2) for both regression models and determine the best fitted model.

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